

F24-085-D-SmartLicense

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Chapter 1 - Introduction

This document outlines the problem and the solution statement, involving all the necessary steps and ideas to tackle this problem to automate the driving license testing system. The current methodologies involve human errors, subjectivity, and many slow processes are involved. As identified during our site visit to the Islamabad traffic police headquarters. Our proposed solution uses sensors, cameras, and AI models to automate the testing process of the driver giving the test, ensuring a more fair, accurate, and leading towards an efficient testing process. The system will look into the driving abilities, including his focus, attention on the road, traffic rules, vehicle handling, and many other aspects.

1.1 Problem Statement

The existing driver's license examination procedure depends significantly on human examiners, resulting in potential biases, inaccuracies, and inefficiencies. Subjective assessments, human error, and variations in testing criteria between assessors are the root causes of these problems. In the tough or reverse test motions, it challenging for examiners to estimate distances precisely, which results in incorrect evaluations. Furthermore, the entire process is prolonged by manual data logging and result creation, making. It requires a lot of resources. In addition, it's also difficult to determine a candidate's attention objectively during their exam, making impartial assessments even more difficult.

To address these issues, an automated system that uses sensor and AI-driven data to provide impartial feedback, eliminate human mistakes, and introduce consistent evaluations is being proposed. As a result, only qualified drivers will be granted licenses, eventually increasing road safety through a more uniform, dependable, and quick testing procedure.

1.2 Scope

This project's scope involves developing and implementing a sensor-integrated, AI-based system to automate driver's license exams. The technology uses cameras and proximity sensors to gather and analyze data to do away with the need for human examiners. The following are some of the system's primary features:

- **Driver performance monitoring:** The system will keep an eye on the driver's compliance with traffic laws, following distance from obstructions, and using mirrors correctly.
- **Parking in reverse and parallel:** The driver's ability to park in both reverse and parallel properly will be assessed by the system. This includes evaluating the driver's control over the car, parking spot location, indication use, and safety precautions including using the mirrors.
- **Processing test data using image processing models:** To guarantee an accurate and impartial assessment of driving prowess, test data will be processed.
- **Automated decision-making:** The system will give the test administrator and the candidate objective results, along with commentary on parking techniques.

- **Web and desktop interfaces:** To facilitate access to and administration of the exam procedure, a web application for applicants and a desktop interface for administrators will be provided.

The objectives of this project are to guarantee scalability for wider deployment, enhance the efficiency, consistency, and fairness of the present testing system and include critical parking skills in the assessment process.

1.3 Modules

The system is broken down into several modules, each of which is in charge of certain functions linked to data collection, processing, interface creation, and system integration. The following modules form the core of the project:

1.3.1 Data Gathering

This module is responsible for collecting data from sensors and cameras installed in the vehicle. The data includes proximity measurements and video footage from both front-facing and interior cameras.

1. Collect data from proximity sensors (distance from objects, obstacles).
2. Capture video from the front-facing camera for traffic sign detection.
3. Capture video from the interior camera for driver behavior analysis.

1.3.2 Data Processing

This module handles cleaning and preparing the raw data collected from sensors and cameras for further analysis.

1. Clean and organize sensor data for processing.
2. Pre-process video frames for efficient model inference.
3. Synchronize sensor and camera data for accurate event correlation.

1.3.3 Model Training

This module involves training AI models, primarily Convolutional Neural Networks (CNNs), to evaluate driving behavior, detect traffic signs, and analyze proximity to obstacles.

1. Train AI models for traffic sign recognition.
2. Develop a model to analyze driver attention behavior.
3. Develop a model to analyze seat-belt usage
4. Develop a model to analyze driver-checking side mirrors.

1.3.4 Output Processing

Once the data is processed and evaluated, this module will handle converting the AI model's predictions into actionable results.

1. Analyze model output to determine if a candidate passes or fails.
2. Provide detailed feedback on specific areas for improvement.
3. Store results in a database for future reference.

1.3.5 Admin Panel Interface Development

The admin panel will be developed as a desktop application in .NET for administrators to manage the system, monitor ongoing tests, and review test results.

1. Build a .NET-based desktop application for administrators.
2. Starting and stopping of driving tests.
3. Monitoring test data and viewing historical records.

1.3.6 User Panel Interface Development

This module allows users to input their CNIC (Citizens National Identity Card) number and view their test results.

1. Display test results along with detailed feedback for the user.

1.3.7 Hardware Integration

This module ensures seamless communication between the hardware (sensors and cameras) and the software system using a Raspberry Pi.

1. Connect proximity sensors to the Raspberry Pi.
2. Connect cameras to the Raspberry Pi.
3. Configure Raspberry Pi to handle data acquisition and transmission.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User Class	Description
ITP Admin	These are employees of the Islamabad Traffic Police who will manage and oversee the automated driver's license testing system. They are responsible for setting up test parameters, monitoring test sessions, and managing the results. The system will be used by approximately 50 administrators across various testing centers. They will require training on how to operate both the software and the hardware components of the system.
Candidates	These are individuals undergoing the driver's license test. They interact with the system during their test, which captures and evaluates their driving performance using sensors and AI. Candidates will receive score-based feedback and results via a user interface. Approximately 1,000 candidates are expected to use the system monthly.
System Developers	A team of software developers responsible for building, deploying, and maintaining the automated testing system. They will work on integrating AI, sensors, and camera functionalities to accurately assess driving abilities. Developers will continuously update the system to incorporate new technologies and improve functionality.

Regulatory Authorities	Government officials ensure that the testing system meets national safety and operation standards for licensing new drivers. They review system reports and audit its operations to certify its accuracy and fairness in evaluating candidates.
Maintenance Technicians	These technicians are tasked with the maintenance and troubleshooting of the hardware used in the testing system, such as sensors, cameras, and computing units. They ensure that the equipment is operational and performs optimally during tests.

Table 1.1: User Class Characteristics

Chapter 2 - Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

2.1.1 Use-case Diagram

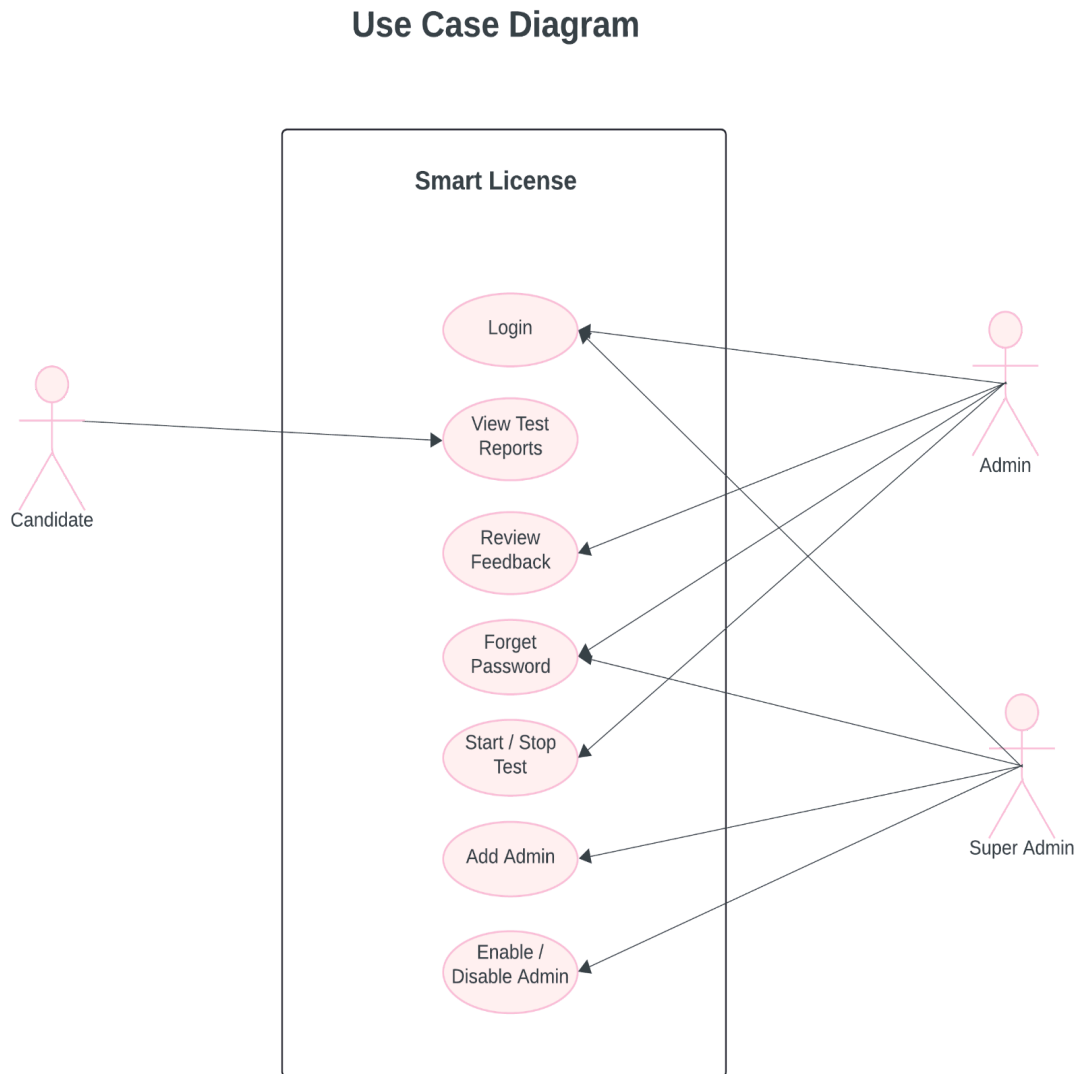


Fig 2.1 : UseCase Diagram

2.1.2 High-Level Use-cases

Use Case # 01: Login

Use Case	Login
Actors	Admin, SuperAdmin
Type	Primary
Description	The admin logs into their management dashboard to access test data and user profiles. The super admin logs into the system to oversee the entire platform operations and perform high-level administrative functions.

Fig 2.1 : UC-01-Login

Use Case # 02: User View Report

Use Case	User View Report
Actors	User
Type	Primary
Description	The user views detailed reports and feedback on their driving tests to understand their performance.

Fig 2.2 : UC-02-User View Report

Use Case # 03: Forgot Password

Use Case	Reset Password
Actors	Admin, Super Admin
Type	Primary
Description	The super admin can request to change their password in case they forget it by getting OTP on email.

Fig 2.3 : UC-03-Forgot Password

Use Case # 04: Add Admin

Use Case	Add Admin
Actors	Super Admin
Type	Primary
Description	The super admin creates admin accounts to which can control and access the system.

Fig 2.4 : UC-04-Add Admin

Use Case # 05: Enable / Disable Admin

Use Case	Enable / Disable Admin
Actors	Super Admin
Type	Primary
Description	The super admin enables or disables admin access to various parts of the system to control permissions and system security.

*Fig 2.5 : UC-05-Enable/Disable Admin***Use Case # 06: Start / Stop Test**

Use Case	Start / Stop Test
Actors	Admin
Type	Primary
Description	The admin starts and stops and oversees the administration of each test.

*Fig 2.6 : UC-06-Start/Stop/Test***Use Case # 07: Review Feedback**

Use Case	Review Feedback
Actors	Admin
Type	Primary
Description	The admin can view and respond to feedback provided by the user on the test report.

Fig 2.7 : UC-07-Review Feedback

2.1.3 Extended Use-cases

Use Case # 01: Login

Attributes	Details
Use Case	Login
Scope	Automated Driving License Testing System
Level	User Goal
Primary Actor	User, Admin, Super Admin
Stakeholders and Interest	User: Secure access to the profile. Admin: Manage user data and schedules.
Preconditions	Users or admins must be registered.
Postconditions	The actor is logged in and directed to the appropriate dashboard.
Main Access Scenario	1. Actor chooses the login. 2. Actor enters credentials. 3. The system verifies and grants access. 4. The actor accesses the dashboard.
Extensions	<i>3a. Actor enters invalid credentials.</i> The system displays an error message indicating the credentials are invalid, allows the actor to retry the login, and provides a "Forgot Password" link if necessary. <i>3b. Actor's account is locked due to multiple failed login attempts.</i> The system temporarily locks the account after a predefined number of failed login attempts, notifies the actor of the lock, and offers a link to initiate the account recovery process. <i>3c. The actor enters credentials but encounters a network connectivity issue.</i> The system detects the network problem, displays an error message, and allows the actor to retry the login once the connection is restored without losing the entered credentials.

Fig 3.1 : ExUC-01-Login

Use Case # 02: User View Report

Attributes	Details
Use Case	View Test Reports
Scope	Automated Driving License Testing System
Level	User Goal

Primary Actor	User, Admin
Stakeholders and Interest	User: View their test results and performance for improvement. Admin: Monitor and verify outcomes.
Preconditions	User or admin must be logged in.
Postconditions	The actor views detailed driving test reports.
Main Access Scenario	1. The actor navigates to the reports section. 2. Actor selects the test. 3. System displays detailed results.
Extensions	<i>2a. Network connectivity is interrupted while loading the test report.</i> The system detects the connectivity loss, displays an error message to the actor, and offers options to retry loading the report or save the request for later retrieval once the connection is restored. <i>2b. Report is unavailable or corrupted.</i> The system notifies the actor of the unavailability or corruption, logs the error for further investigation, and provides an option to request a re-generation of the report.

Fig 3.2 : ExUC-02-User View Report

Use Case # 03: Forgot Password

Attributes	Details
Use Case	Forgot Password
Scope	Automated Driving License Testing System
Level	System Operation
Primary Actor	Admin, Super Admin
Stakeholders and Interest	Super Admin: Needs to reset their password to regain access to the system. Admin: Needs to reset their password to regain access to the system. System: Must ensure a secure process for password reset using OTP to avoid unauthorized access.
Preconditions	Super Admin / Admin must have a valid account, and the email address must be registered in the system.
Postconditions	A new password is set for the Admin / Super Admin after successful OTP verification.
Main Access Scenario	1. The Admin / Super Admin navigates to the "Forgot Password" page. 2. The system prompts the Admin / Super Admin to enter their registered email address.

	3. The Admin / Super Admin submits the email address. 4. The system validates the email and sends a one-time password (OTP) to the provided email. 5. The Admin / Super Admin receives the OTP and enters it into the system. 6. The system verifies the OTP. 7. The system prompts the Admin / Super Admin to enter a new password and confirm it. 8. The Admin / Super Admin enters the new password. 9. The system confirms that the password reset is successful and notifies the Admin / Super Admin
Extensions	<i>4a: The email address entered is not registered.</i> The system notifies the Admin / Super Admin that the email is not found in the system, prompting them to enter a valid email address. <i>5a: The Admin / Super Admin does not receive the OTP.</i> The system provides an option to resend the OTP or instructs the Admin / Super Admin to check spam/junk folders. <i>6a: The OTP entered is incorrect or expired.</i> The system notifies the Admin / Super Admin of the incorrect/expired OTP and prompts them to re-enter or request a new OTP. <i>3a: Network connectivity is lost during the OTP verification.</i> The system detects the disconnection, alerts the Admin / Super Admin, and allows them to retry once the connection is restored.

Fig 3.3 : ExUC-03-Forgot Password

Use Case # 04: Add Admin

Attributes	Details
Use Case	Add Admin
Scope	Automated Driving License Testing System
Level	System Operation
Primary Actor	Super Admin
Stakeholders and Interest	Super Admin: Needs to efficiently manage administrative roles by adding new admins to the system. Admins: Newly added admins require appropriate access to perform their roles effectively.
Preconditions	Super Admin must be logged in with rights to modify access levels.
Postconditions	A new admin is successfully added to the system with the appropriate access level and permissions.

Main Access Scenario	1. Super Admin accesses the admin management interface. 2. Super Admin enters the necessary information for the new admin (e.g., name, email). 3. System validates the provided information e.g. format. 4. System creates a new admin account and assigns the specified roles and permissions. 5. The system confirms the addition and notifies the Super Admin.
Extensions	<i>3a. Super Admin enters incomplete or invalid information for the new admin.</i> The system detects the error, alerts the Super Admin to correct the information, and prevents proceeding until all fields are properly completed. <i>4a. The system finds an existing admin with the same email or username.</i> The system notifies the Super Admin of the duplicate information, suggests altering the username or email, and allows the Super Admin to retry the process with unique credentials. <i>3a. Network connectivity is lost during the Add Admin process.</i> The system detects the disconnection, alerts the Super Admin of the connectivity issue, and allows them to retry once the connection is restored.

Fig 3.4 : ExUC-04-Add Admin

Use Case # 05: Enable / Disable Admin

Attributes	Details
Use Case	Enable / Disable Admin
Scope	Automated Driving License Testing System
Level	System Operation
Primary Actor	Super Admin
Stakeholders and Interest	Super Admin: Efficient management of admin roles. Admin: Relies on clear and functional administrative privileges.
Preconditions	Admin accounts are updated as needed (created, Enabled, Disabled, or deleted).
Postconditions	The system updates the Admin profile and their privileges to the account.
Main Access Scenario	1. Super Admin accesses the admin management section. 2. Super Admin performs necessary changes to admin accounts. 3. System updates accounts and privileges accordingly.
Extensions	<i>2a. Attempt to delete an admin who is currently conducting a test.</i> The system prevents the deletion of the admin, notifies the Super Admin of the ongoing test, and suggests rescheduling the deletion until the test is completed.

	<i>2b. Super Admin attempts to disable an admin who is actively logged in.</i> The system alerts the Super Admin about the active session, offers the option to force logout the admin, and proceeds with disabling the account only after confirmation to maintain system integrity. <i>2c. Network connectivity is lost during the admin account update process.</i> The system detects the disconnection, alerts the Super Admin of the connectivity issue, and allows them to retry the update once the connection is restored to ensure data consistency. <i>2d. Data corruption occurs while updating an admin's privileges.</i> The system identifies the data corruption, automatically rolls back to the previous stable state, notifies the Super Admin of the issue, and initiates a data integrity check to prevent future occurrences.
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Fig 3.5 : ExUC-05-Enable/Disable Admin

Use Case # 06: Start / Stop Test

Attributes	Details
Use Case	Manage Test
Scope	Automated Driving License Testing System
Level	Admin Task
Primary Actor	Admin
Stakeholders and Interest	Admin: Needs to efficiently manage the test, ensuring reliable scheduling and starting/stopping test events. User: Requires accurate and timely test management and notification.
Preconditions	Admin must be logged into the system with the correct permissions to manage tests.
Postconditions	Test schedules are updated in the system, and notifications are sent to the users involved.
Main Access Scenario	1. The Admin logs into the system. 2. The Admin navigates to the test management interface by selecting user. 3. The Admin selects the test to start or stop. 4. The Admin initiates the start/stop command for the test. 5. The system validates the test details and current status. 6. The system updates the test status and records the time of the action. 7. The system notifies the relevant users (including the test taker) about the test status update. 8. The system generates a confirmation for the Admin regarding the successful operation.

Extensions	<i>2a: The Admin attempts to start/stop a test that is already in progress or stopped.</i> The system alerts the Admin of the current status and prevents duplicate operations. <i>5a: The Admin attempts to start a test for a user who is not eligible.</i> The system checks user eligibility and notifies the Admin with an error message, prompting them to review user details. <i>6a: The network disconnects during the test operation.</i> The system detects the interruption, saves progress locally, and prompts the Admin to retry once the connection is restored.
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Fig 3.6 : ExUC-06-Start/Stop Test

Use Case # 07: Review Feedback

Attributes	Details
Use Case	Review Feedback
Scope	Automated Driving License Testing System
Level	Admin Task
Primary Actor	Admin
Stakeholders and Interest	Admin: Needs to review user feedback to improve the system and respond to user issues. User: Requires timely responses and resolutions for their feedback on the test reports.
Preconditions	The admin is logged in with proper permissions to access the feedback section.
Postconditions	The feedback is reviewed, responded to, and stored in the system with proper resolution logs.
Main Access Scenario	1. The admin logs into the system. 2. The admin navigates to the feedback section on their dashboard. 3. The system retrieves and displays feedback provided by users on test reports. 4. The admin selects specific feedback to review. 5. The admin assesses the feedback, analyzes the issue, and decides on an appropriate response. 6. The admin responds to the feedback through the system interface. 7. The system records the response, sends the reply to the user, and stores the interaction in the database. 8. The system confirms the feedback has been addressed and marks it as

	reviewed.
Extensions	7a: Network connectivity is lost while responding to feedback. The system saves the admin's response locally and prompts them to retry once the connection is restored

Fig 3.7 : ExUC-07-Review Feedback

2.2 Functional Requirements

This section presents the functional requirements. Each requirement addresses a specific feature or system function necessary for the automation of the driver's license testing process.

2.2.1 Data Gathering Module

- **FR1.1:** The system shall collect data from proximity sensors installed at strategic locations around the vehicle i.e., two in front, two in the center, and two at the back of the car, all attached one on the left and one on the right side of the car, to measure exact distances to cones during specific parking maneuvers and while navigating through the testing course.
- **FR1.2:** The system shall capture high-resolution video footage from a front-facing camera mounted on the dashboard to monitor compliance with traffic signs and cones. This data capture will include:
 - Camera angle and field of view adjustments.
 - Data compression and storage methods.
 - Timestamping and synchronization with other data sources.
- **FR1.3:** The system shall record video from interior cameras focused on the driver to assess compliance with driving protocols such as seatbelt usage, driver's attention, and hand position on the steering wheel. This will involve:
 - Interior camera positioning and coverage area.
 - Behavioral markers for evaluation (e.g., head turns, eye movement).
 - Secure storage and access control measures for video data.

2.2.2 Data Processing Module

- **FR2.1:** The system shall clean, preprocess, and standardize data collected from all onboard sensors and cameras to ensure the data is ready for analysis. This includes:
 - Noise reduction algorithms.
 - Data normalization procedures.
 - Missing data handling.
 - Timestamp correlation for data synchronization.
 - Preparation of datasets for analysis and reporting.
- **FR2.2:** The system shall synchronize data from multiple sensors and cameras for a cohesive analysis of driving performance. This synchronization will ensure:
 - Alignment of video with sensory data.
 - Integration of data for complex maneuver analysis.
- **FR2.3:** The system shall preprocess video frames to ensure smooth image analysis for traffic sign detection and driving behavior evaluation, involving:

- Image optimization for processing.
- Ability to process video streams.

2.2.3 Model Training and Evaluation Module

- **FR3.1:** The system shall use image processing techniques and object detection machine learning models to observe driver behavior such as seatbelt adherence, hand positioning on the wheel, and driver attention including:
 - Training on datasets comprising various driving conditions.
 - Checking the model's performance metrics.
- **FR3.2:** The system shall utilize object detection ML models for traffic sign detection and analyze driver responses to traffic signs. This functionality will cover:
 - Model training with a diverse set of traffic sign images.
- **FR3.3:** The system shall evaluate the candidate for parking performance using sensor and video data to assess the driver's ability to drive the vehicle and to park it correctly. Assessments will include:
 - Parallel and reverse parking evaluations.
 - Assessment of control smoothness and precision.
 - Corrective feedback to the driver.

2.2.4 Output Processing Module

- **FR4.1:** Based on a predetermined set of criteria and model evaluations, the system will automatically determine whether the candidate passed or failed the driving test. This process will involve:
 - Defining pass/fail criteria.
 - Automated score generation.
 - Result computation.
- **FR4.2:** For each candidate, the system shall generate a detailed feedback report indicating their performance on specific driving tasks such as parking and adherence to traffic rules. Reports will include:
 - Personalized feedback based on performance data.
 - Details of key performance indicators.
 - Recommendations for improvements.

2.2.5 User Interface Module

- **FR5.1:** The user shall be able to view a detailed report of the test along with feedback on the web-panel. This requirement will encompass:

- Responsive and user-friendly design.
- User-friendly navigation and accessible design for all user types.
- **FR5.2:** The user shall be able to submit feedback on their test report through an easy-to-use form. This functionality will include:
 - A dedicated feedback section.
 - Confirmation message for successful feedback submission.
 - Storage of submitted feedback for further reviews by administrators.

2.2.6 Admin Panel Interface Module

- **FR6.1:** The system shall provide a robust, .NET-based desktop application for administrators to efficiently manage ongoing tests, monitor data from sensors and cameras, and adjust test settings. This includes:
 - Dashboard displaying ongoing tests and status.
- **FR6.2:** The system shall enable administrators to respond to user feedback submitted on test reports. This functionality will include:
 - A feedback review section where all submitted feedback is displayed.
 - Tools for administrators to reply directly to user feedback.
 - Option to mark feedback as resolved once addressed.

2.2.7 Hardware Integration Module

- **FR7.1:** The system shall integrate proximity sensors, cameras, and the Raspberry Pi to capture and transmit data during the test. This will include:
 - Basic procedures for initializing hardware before each test.
 - Simple error detection and notification for hardware issues.
 - Standard data transmission methods to maintain data integrity.
- **FR7.2:** The system shall manage communication between the hardware components (sensors and cameras) and the central server. This will include:
 - Collecting data from the hardware during the test.
 - Transmitting data to the server for processing.

2.3 Non-Functional Requirements

The following are the non-functional requirements of the system:

2.3.1 Reliability

RLB-1: The system shall automatically log errors and failures for review without requiring manual intervention.

2.3.2 Usability

USE-1: The system interface shall allow administrators to configure and control tests with no more than 3 steps.

USE-2: Drivers shall receive their test results within 10 minutes after test completion.

USE-3: The user interface shall allow users to be able to view test reports with no more than 2 steps.

2.3.3 Performance

PER-1: The system shall process sensor and camera data and generate test results within 10 minutes.

2.3.4 Security

SEC-1: The system will use role-based authorization to ensure that only authorized users can access the candidate data and initiate tests.

SEC-2: The system will prevent any unauthorized users from accessing the system without permissions at the administrator level.

2.4 Domain Model

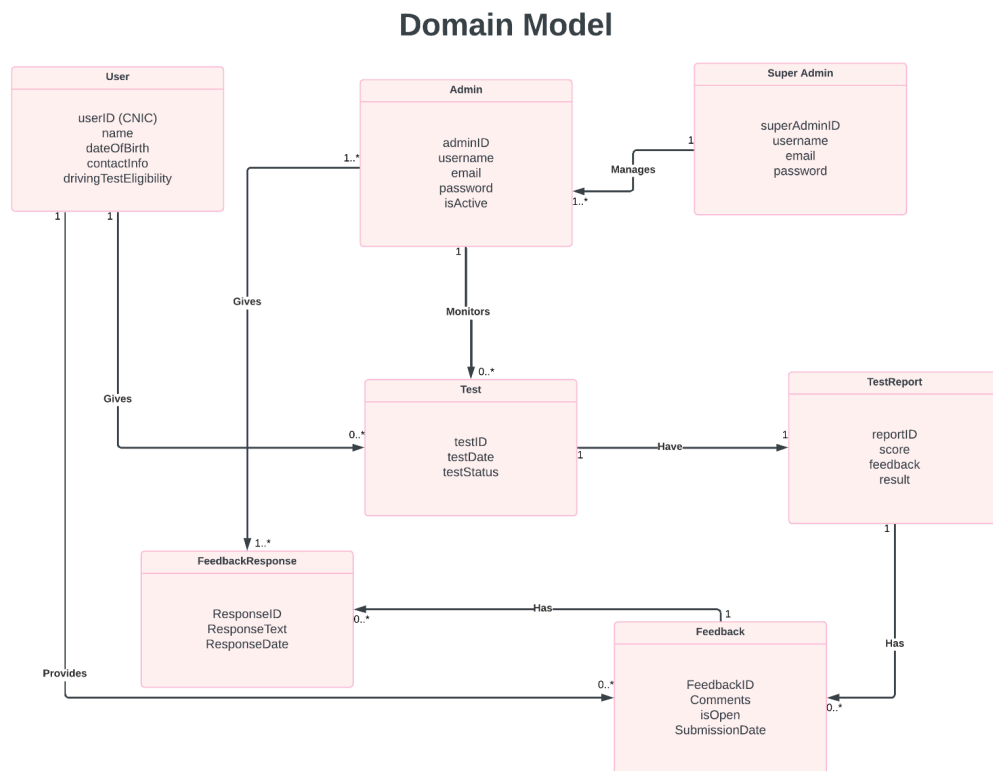


Fig 2.2 : Domain Model Diagram

Chapter 3 - System Overview

The Smart Licensing System streamlines the process of testing for driver's licenses by using proximity sensors and AI models for monitoring and analysis. This approach improves both accuracy and fairness by gathering and evaluating data on driving behavior and performance, leading to objective, fact-based outcomes. Therefore this system minimizes human error, guarantees uniform assessments, and delivers comprehensive feedback to both candidates and system managers.

3.1 Architectural Design

This section outlines the architectural design of the driver's license test automation system, which consists of several distinct modules working together to automate the testing process. The system is organized into modules focused on *data gathering*, *data processing*, *model evaluation*, *output processing*, *user interface*, and an *admin panel*. These modules do everything from collecting data via sensors and cameras attached to the vehicle, to processing and evaluating this data from these hardware to provide test results and reports.

The system will be built on a multi-tiered architecture to enhance manageability and scalability. This structure separates the system into layers for *data management*, *application logic*, and *user interaction*, which simplifies maintenance and allows for scalability if required in the future. The **presentation layer** includes user interfaces for candidates and admins, the **application layer** handles the data processing of data from the hardware and model evaluation, and **the data layer** manages storage.

This architecture ensures each component can be updated independently without disturbing the system's overall functionality.

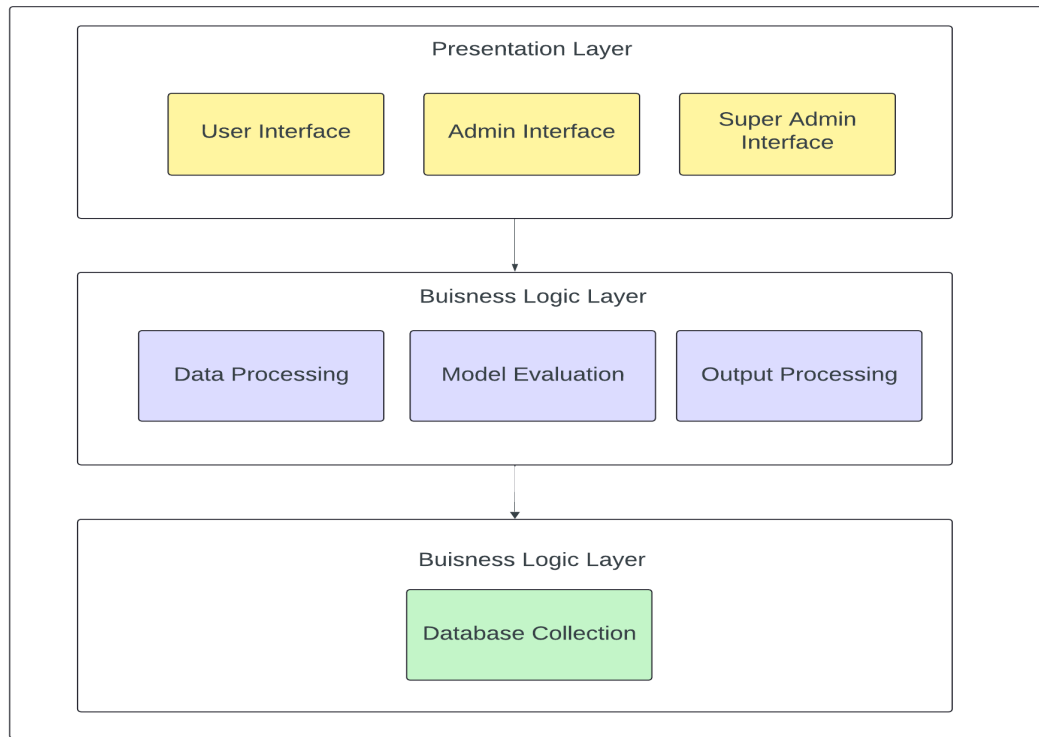


Fig 3.1 : Architecture Design of the system

3.2 Design Models

3.2.1 Activity Diagrams:

Use Case # 01: Login

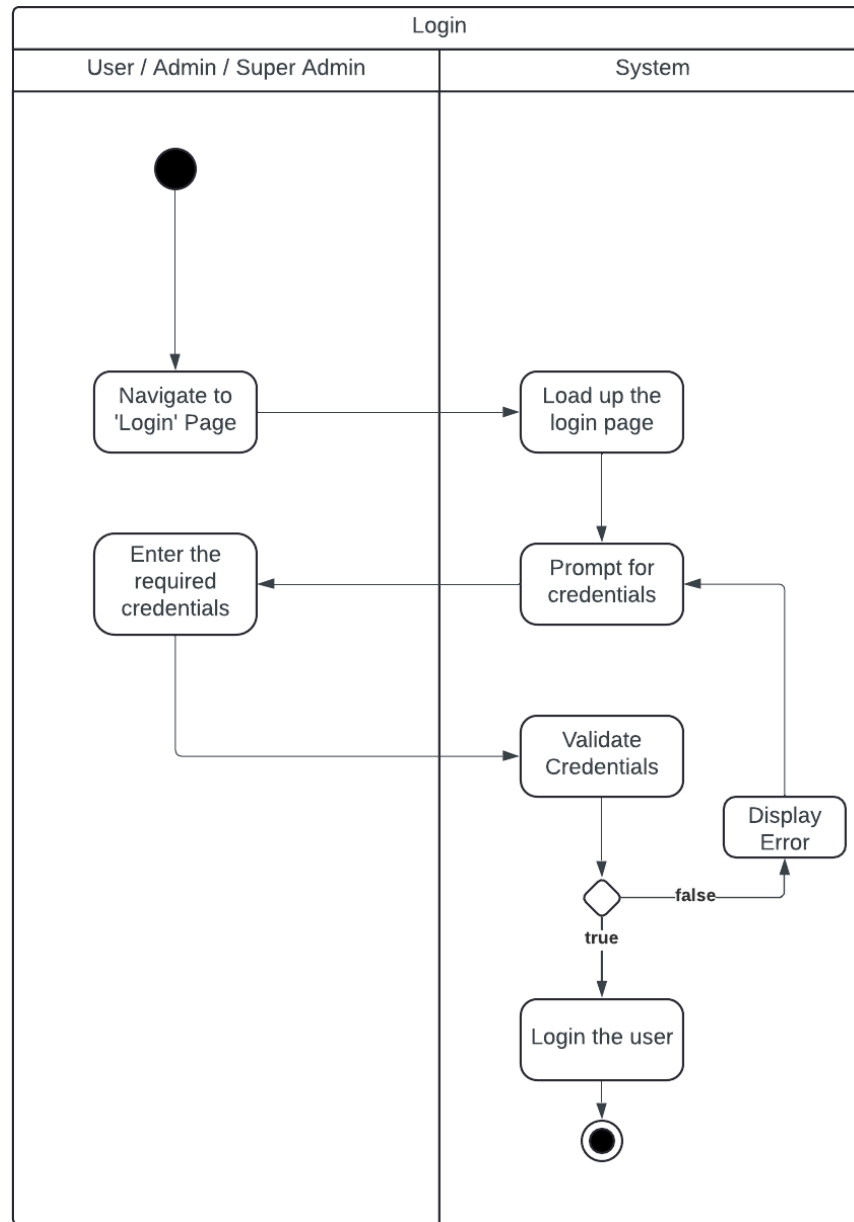


Fig 3.2.1 : Login Activity Diagram

Use Case # 02: User View Report

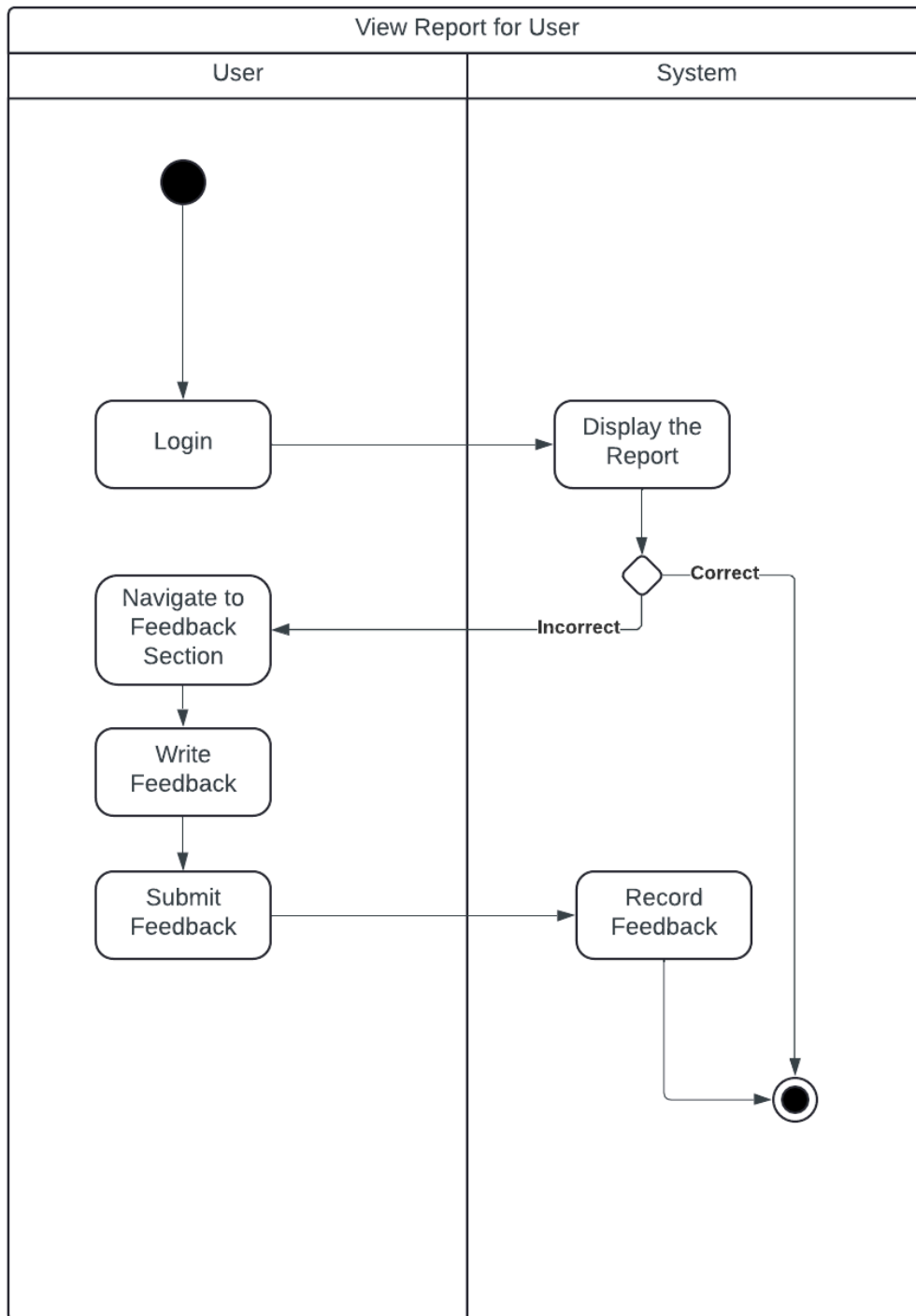


Fig 3.2.2 : User View Report Activity Diagram

Use Case # 03: Forget Password

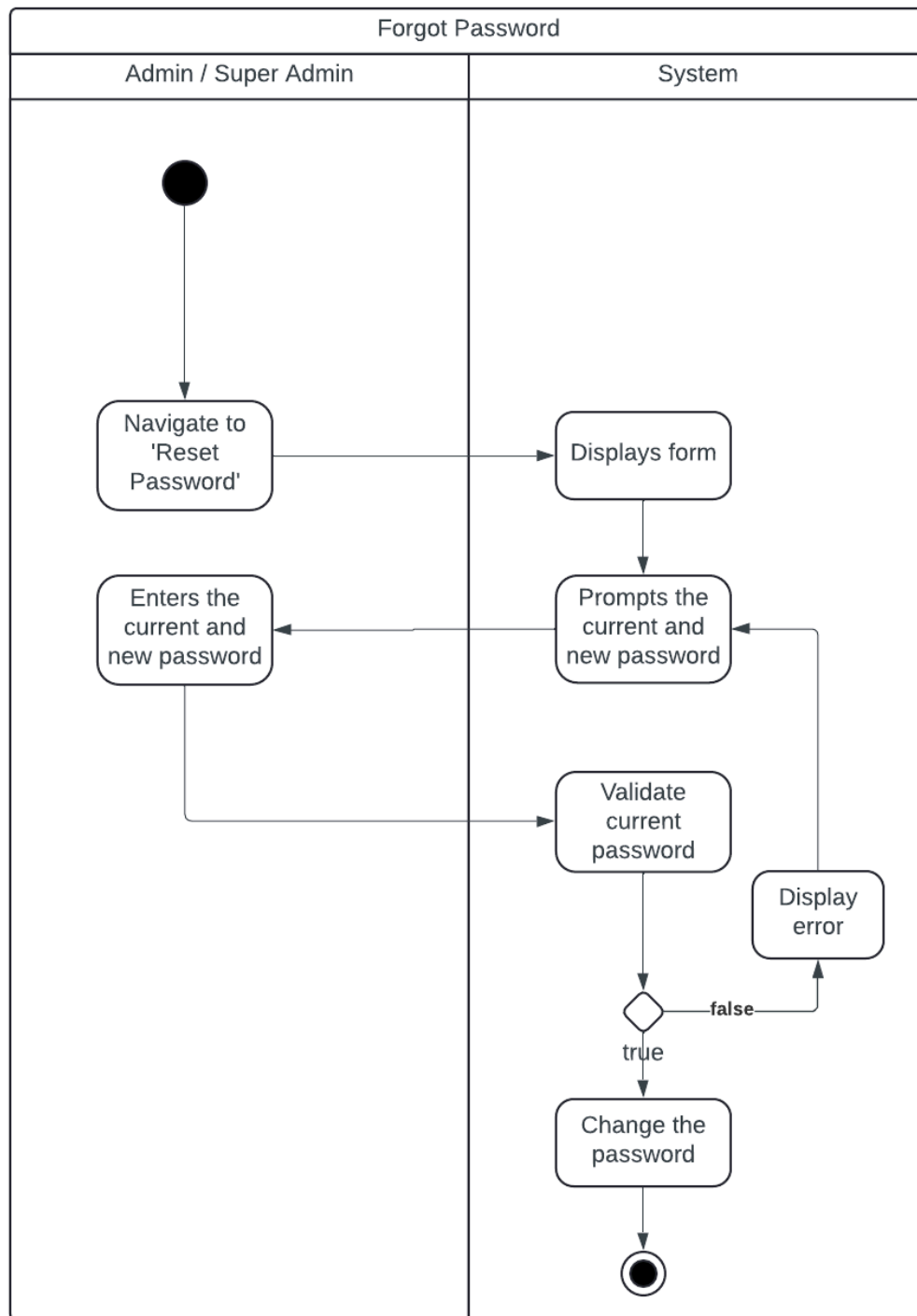


Fig 3.2.3 : Forget Password Activity Diagram

Use Case # 04: Add Admin

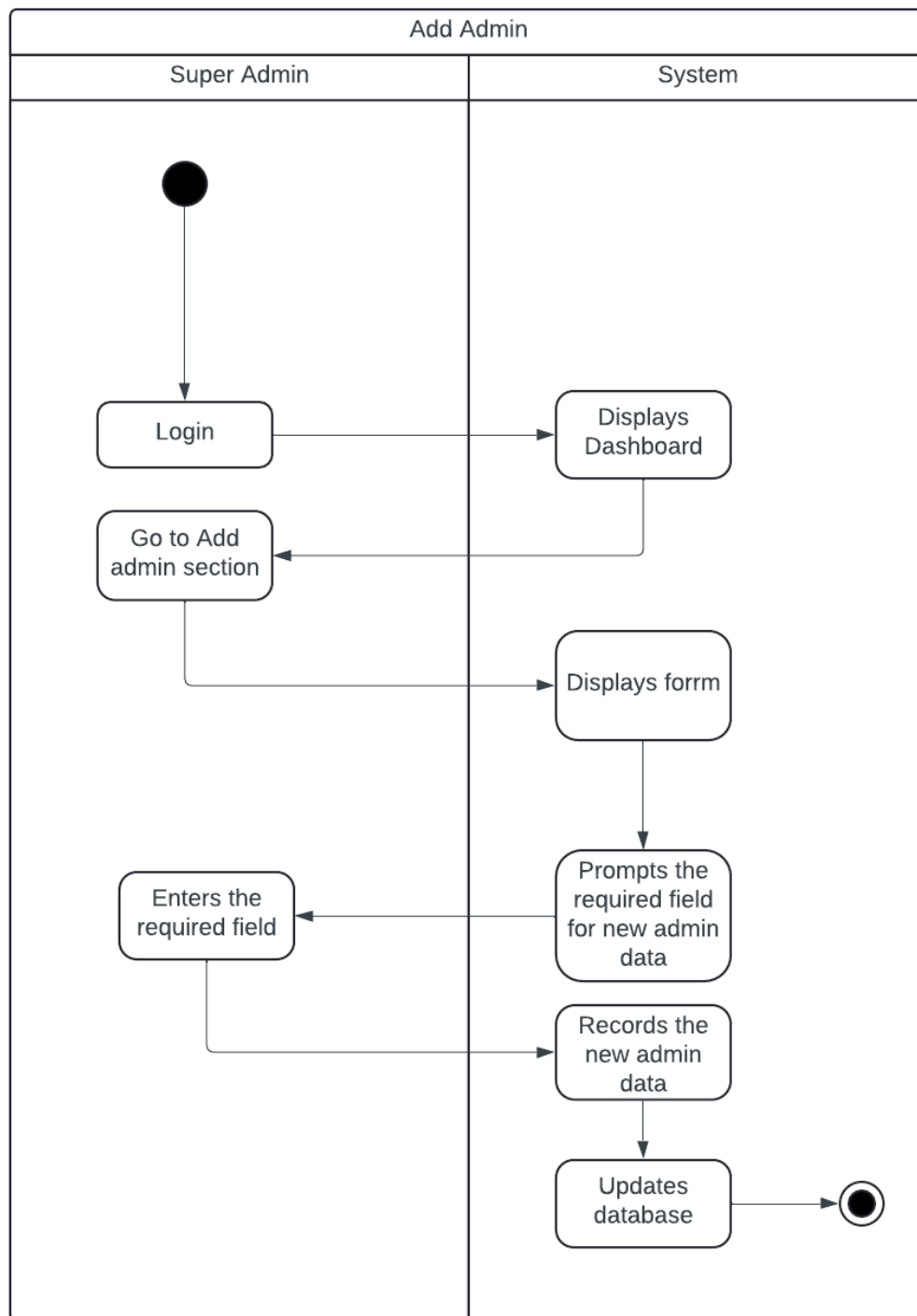


Fig 3.2.4 : Add Admin Activity Diagram

Use Case # 05: Enable/Disable Admin

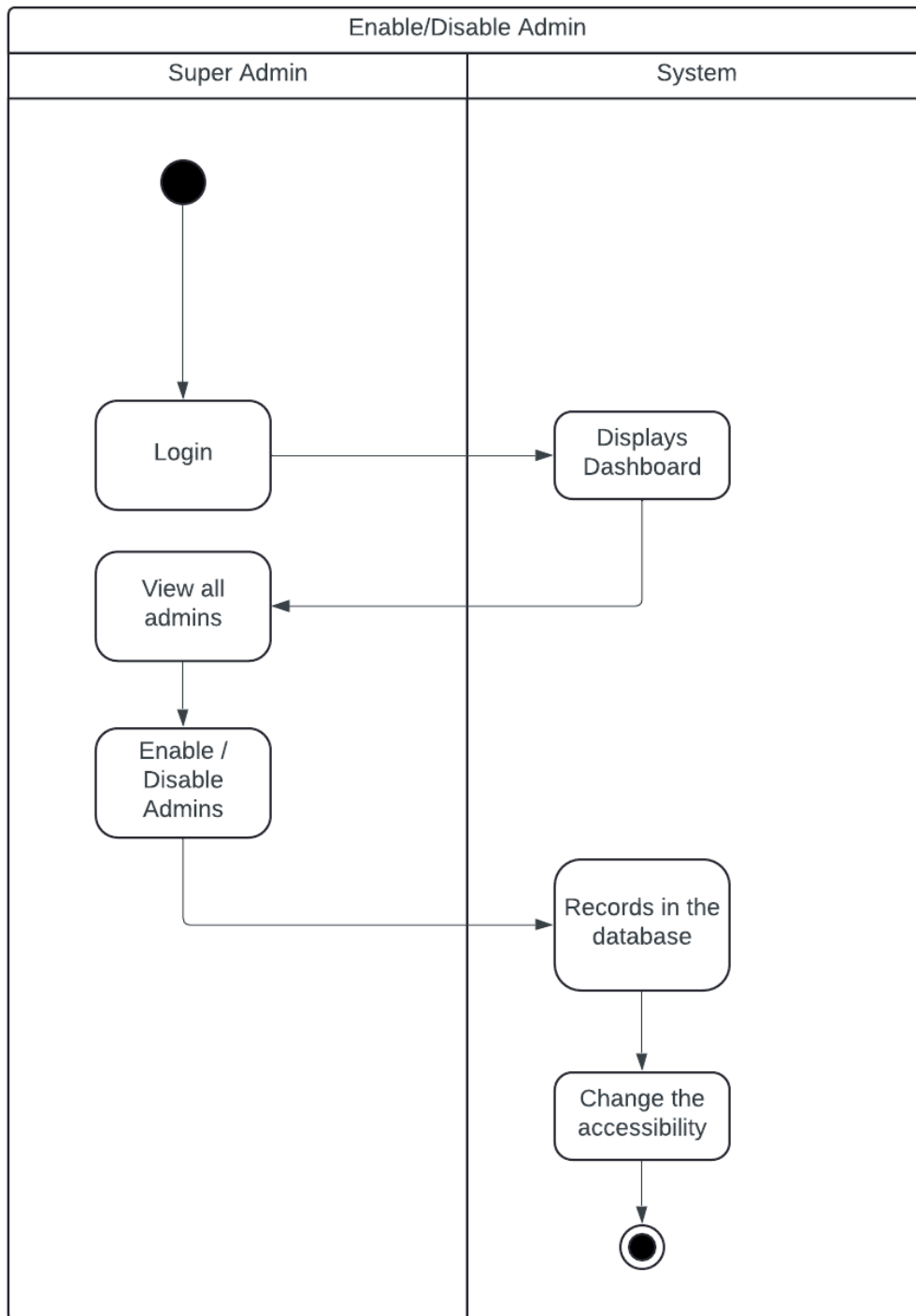


Fig 3.2.5 : Enable/Disable Admin Activity Diagram

Use Case # 06: Start / Stop Test

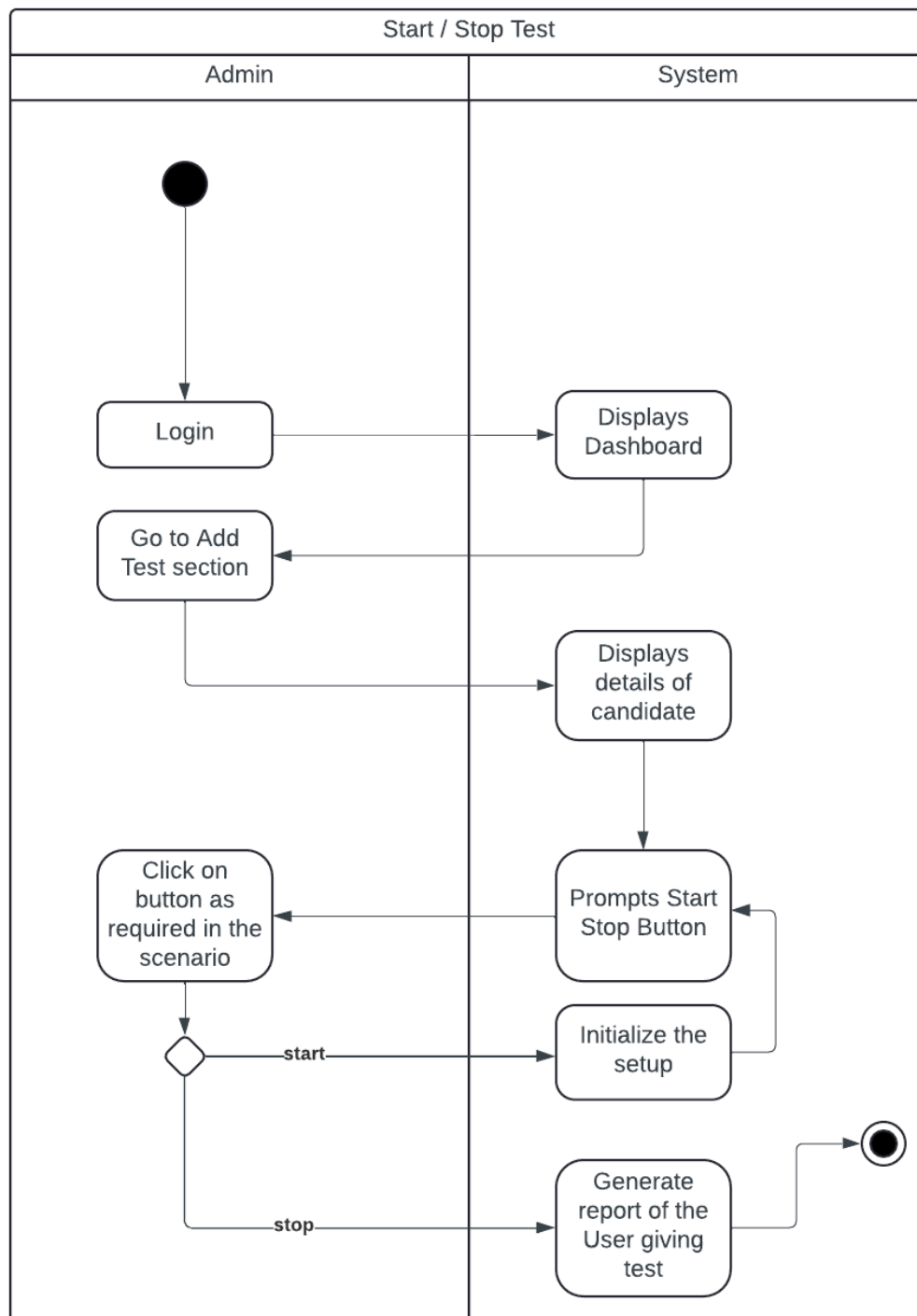


Fig 3.2.6 : Start/Stop Test Activity Diagram

Use Case # 07: Admin Review Feedback

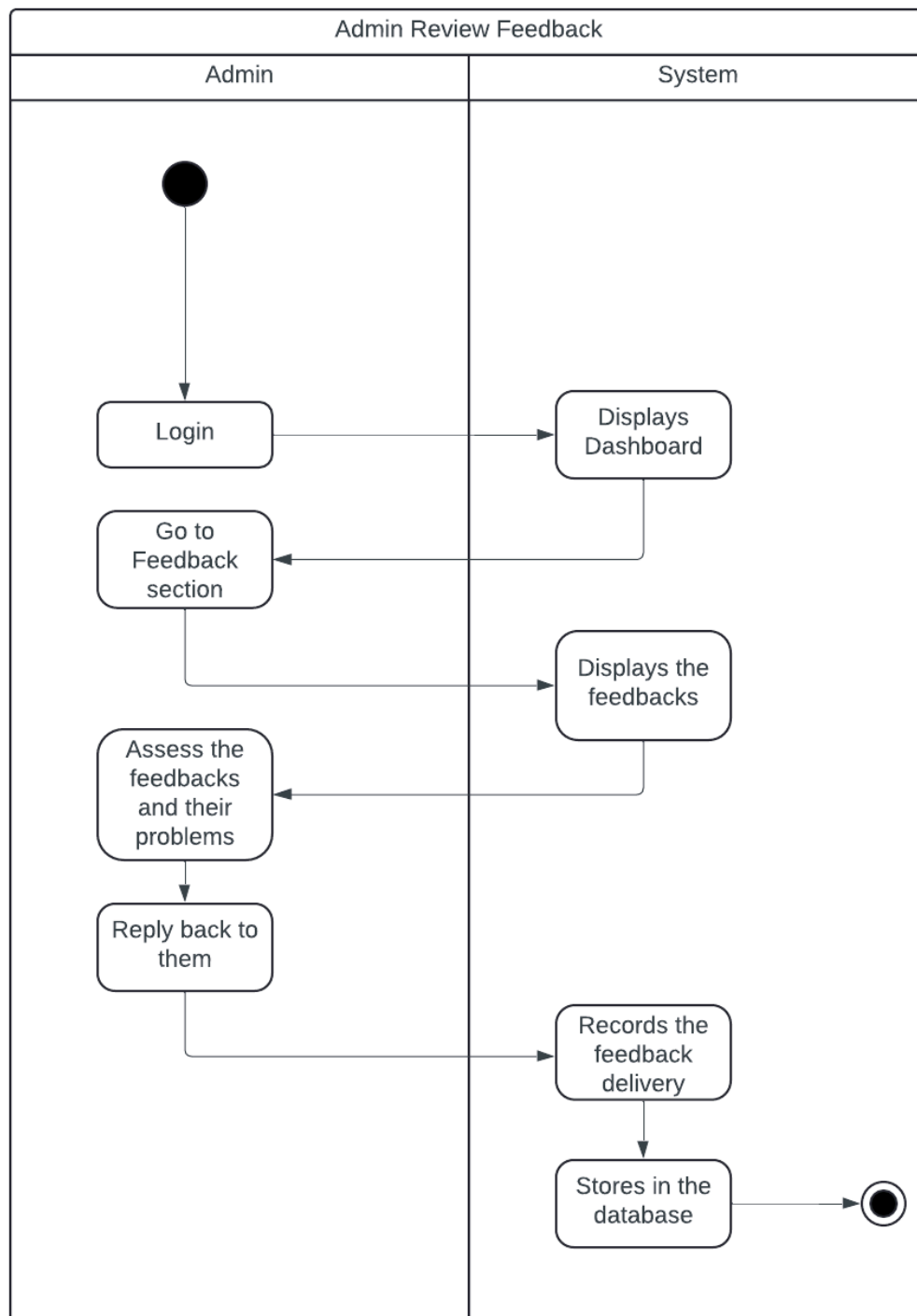


Fig 3.2.7 : Add Review Feedback Activity Diagram

3.2.2 System Sequence Diagrams:

Use Case # 01: Login

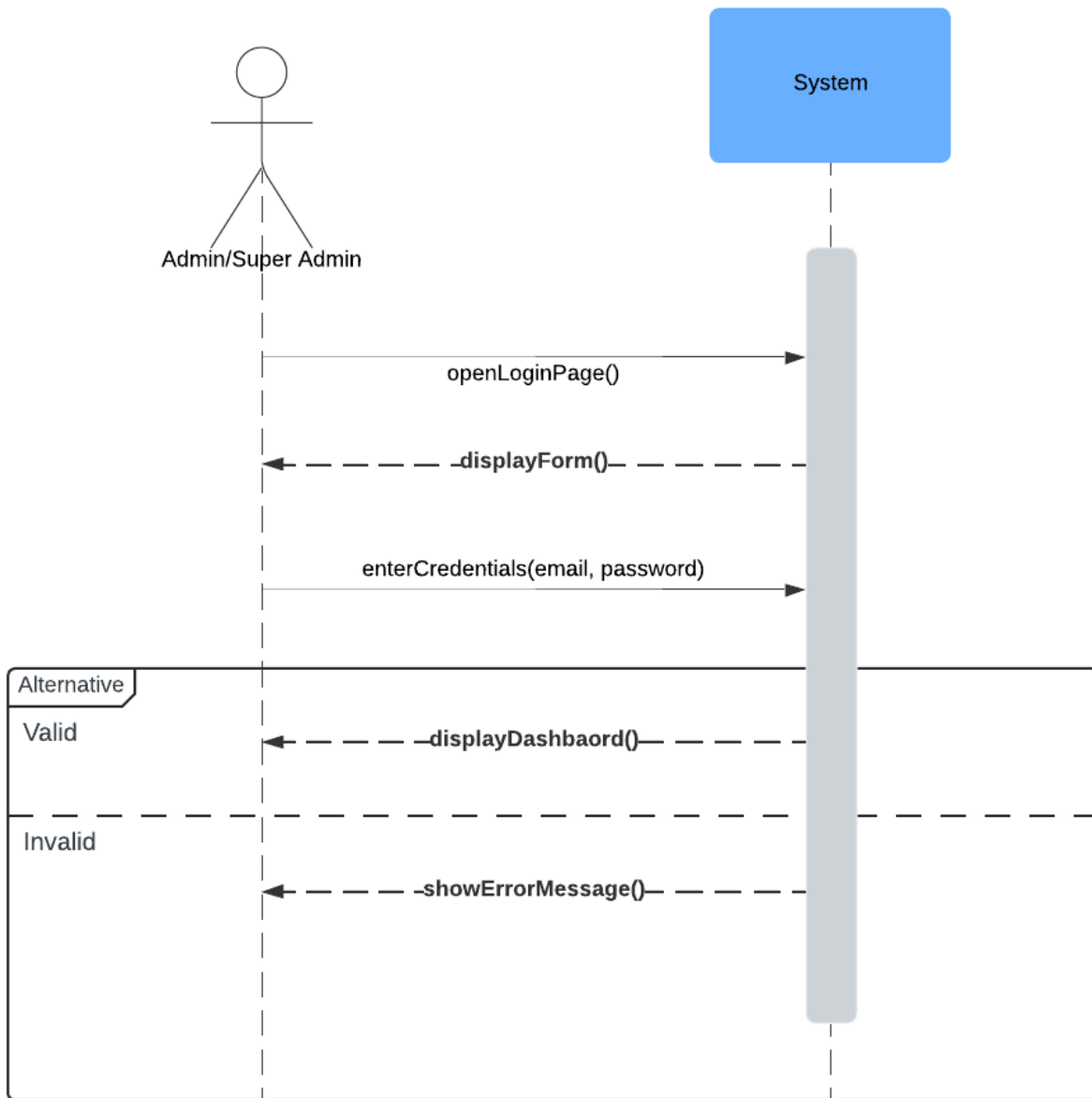


Fig 3.3.1 : Login System-level Sequence Diagram

Use Case # 02: User View Report

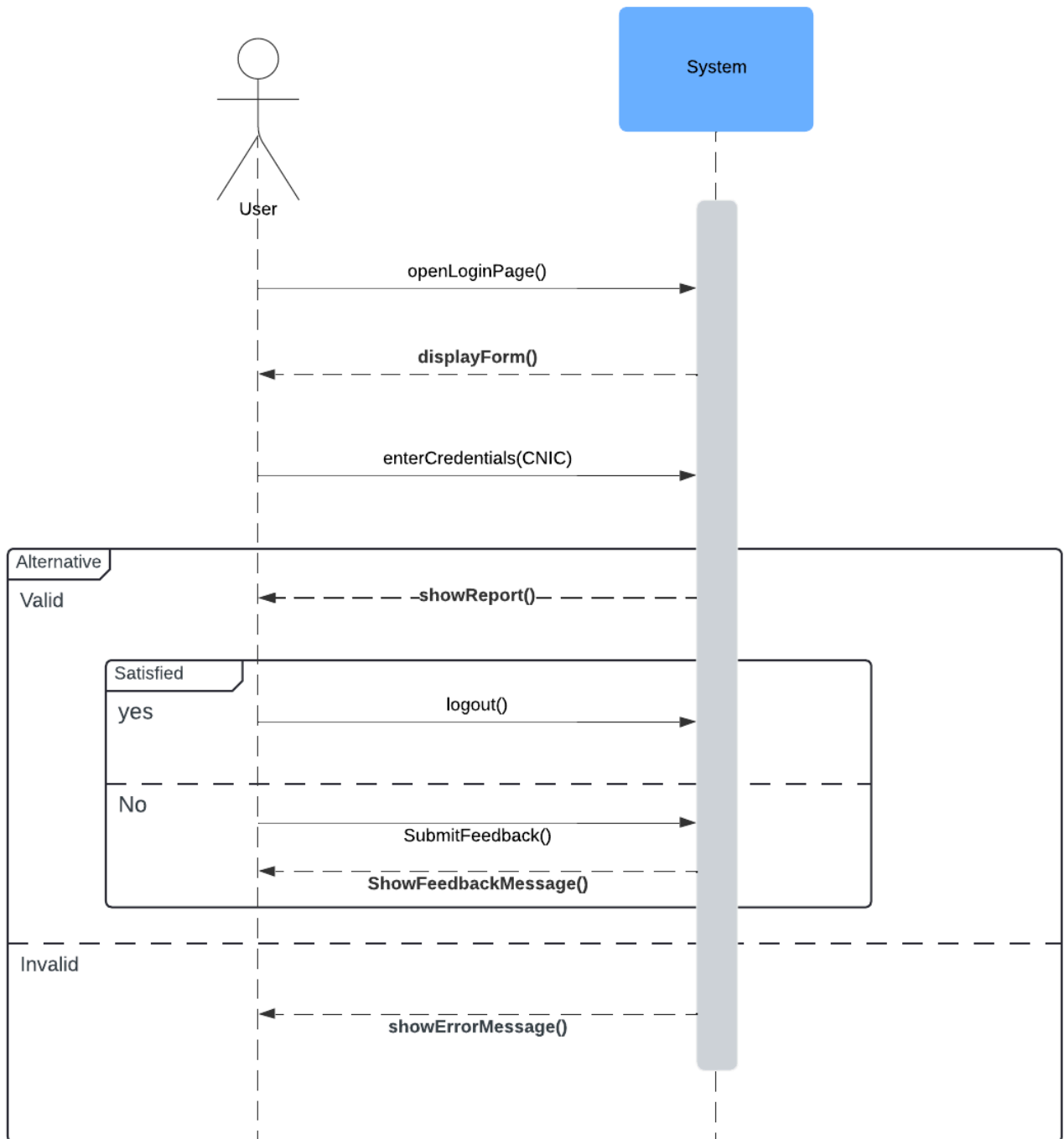


Fig 3.3.2 :User View Report System-level Sequence Diagram

Use Case # 03: Forgot Password

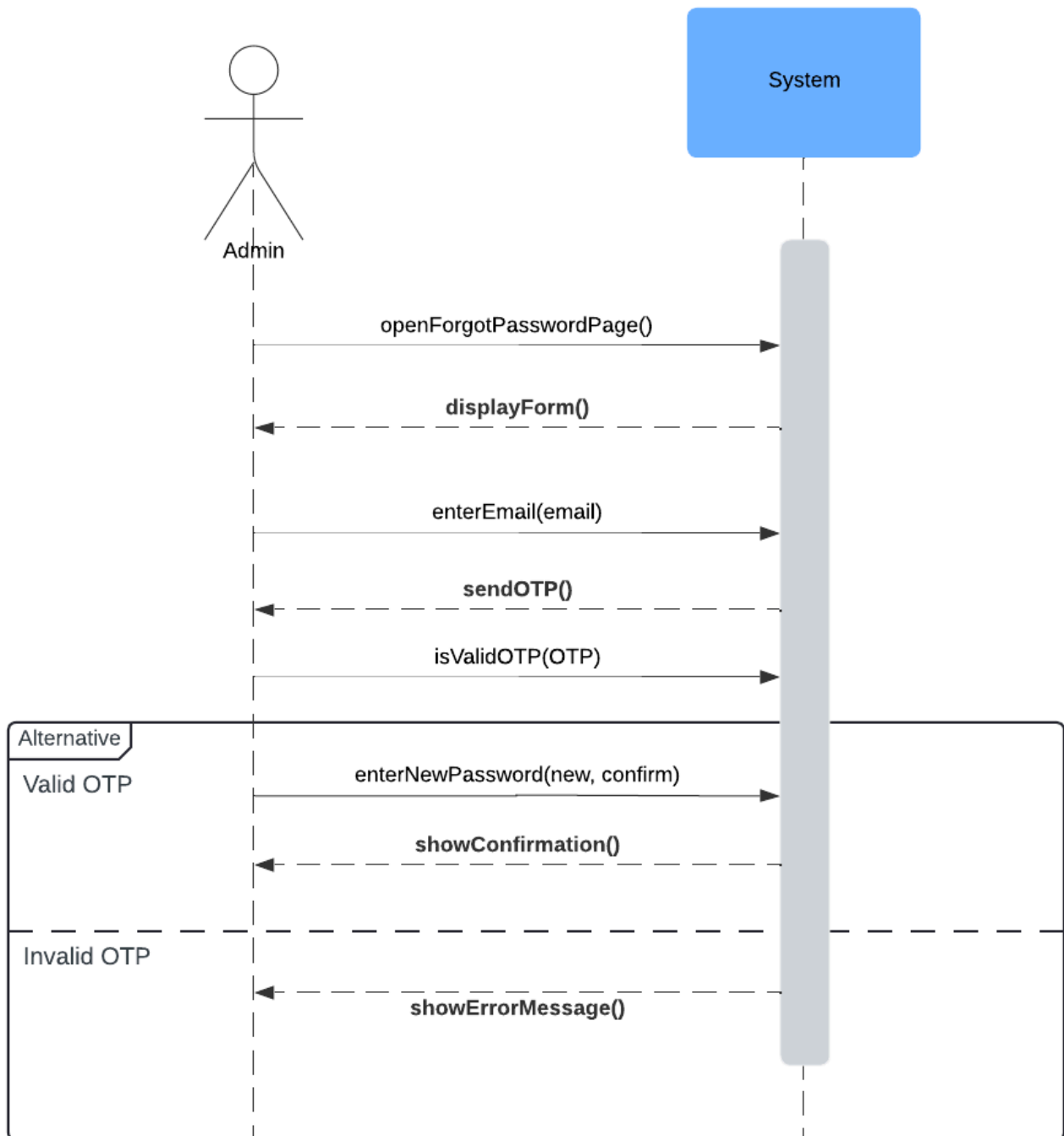


Fig 3.3.3 : Forgot Password System-level Sequence Diagram

Use Case # 04: Add Admin

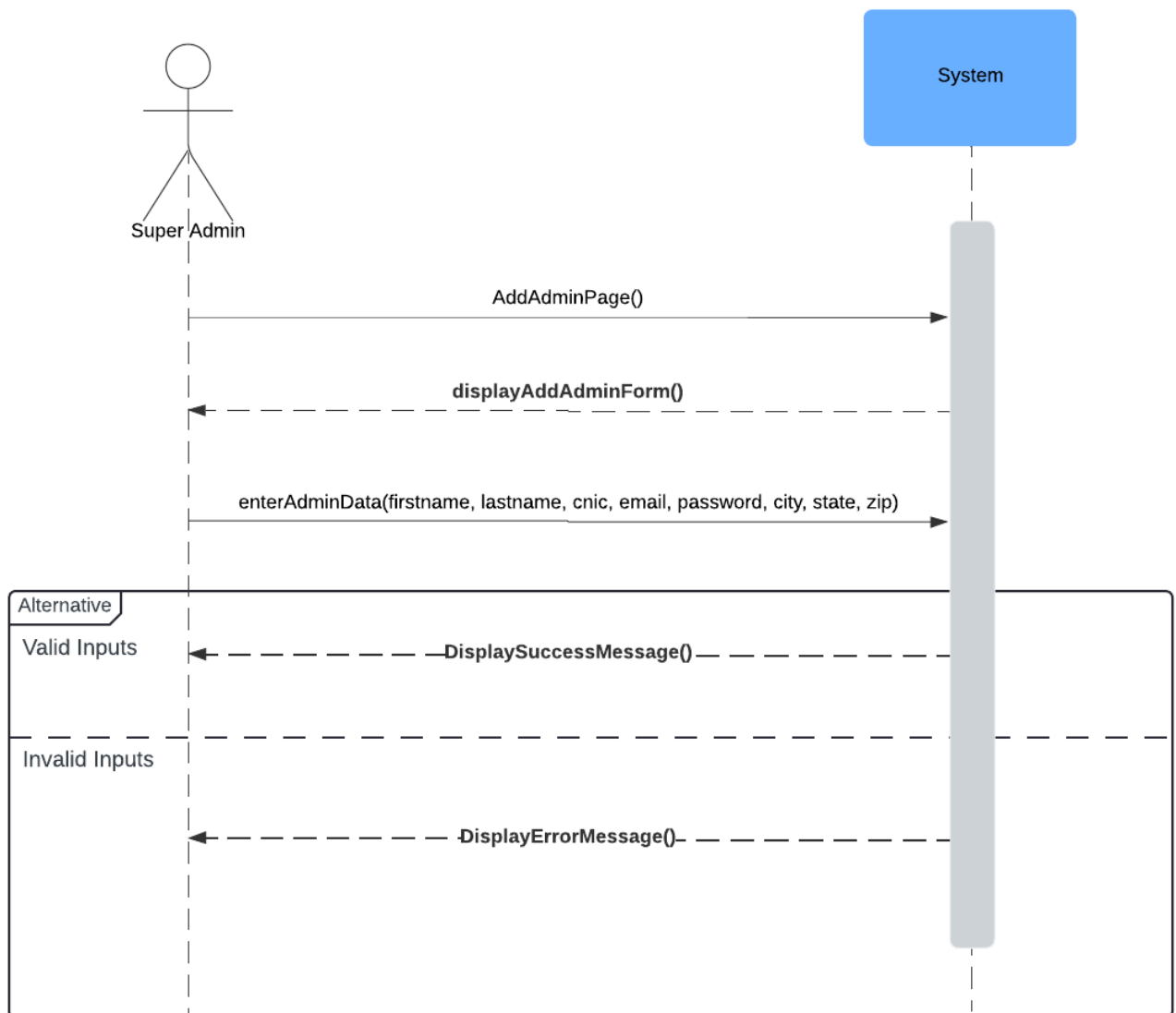


Fig 3.3.4: Add Admin System-level Sequence Diagram

Use Case # 05: Enable / Disable Admin

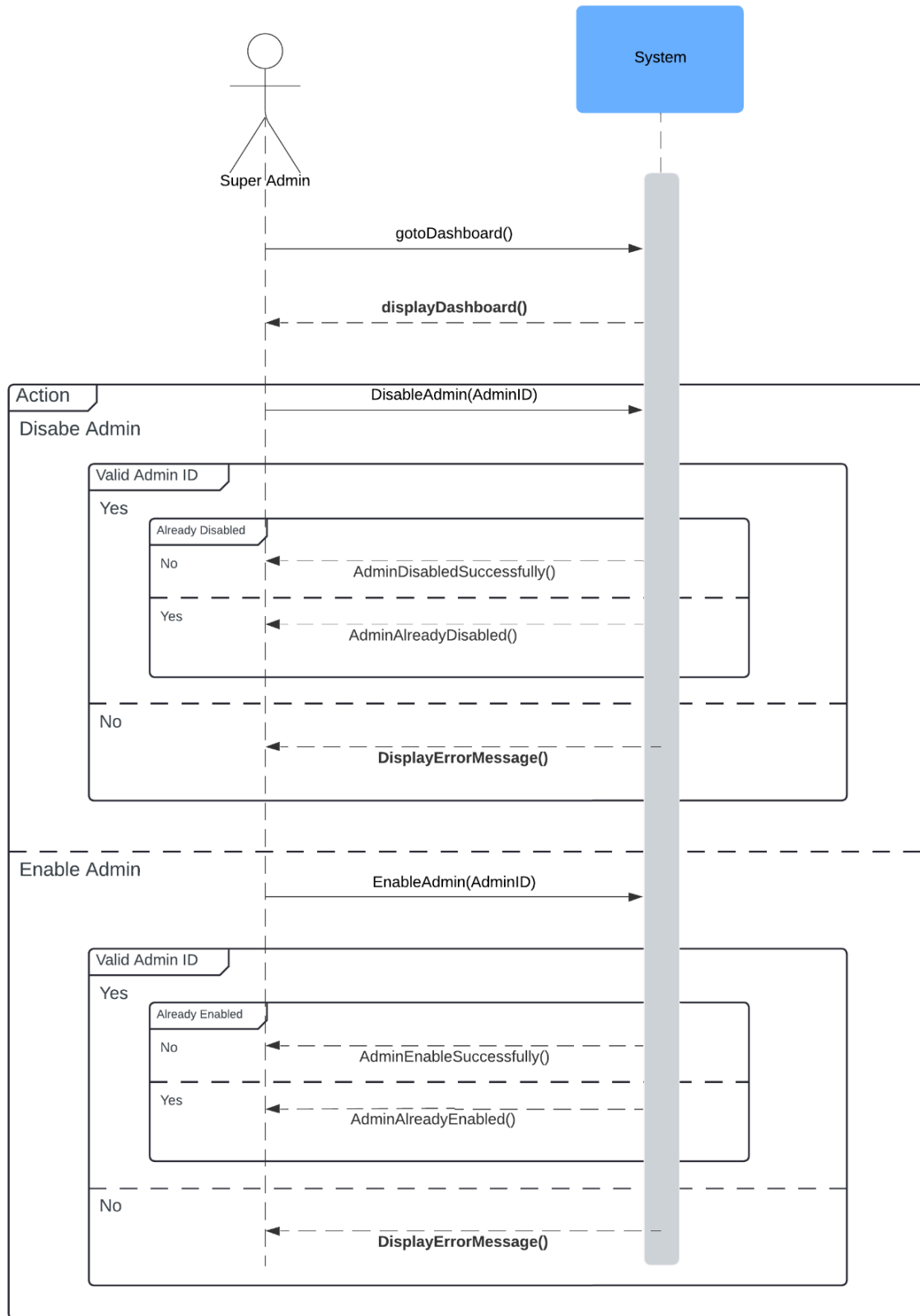


Fig 3.3.5 :Enable/Disable Admin System-level Sequence Diagram

Use Case # 06: Start / Stop Test

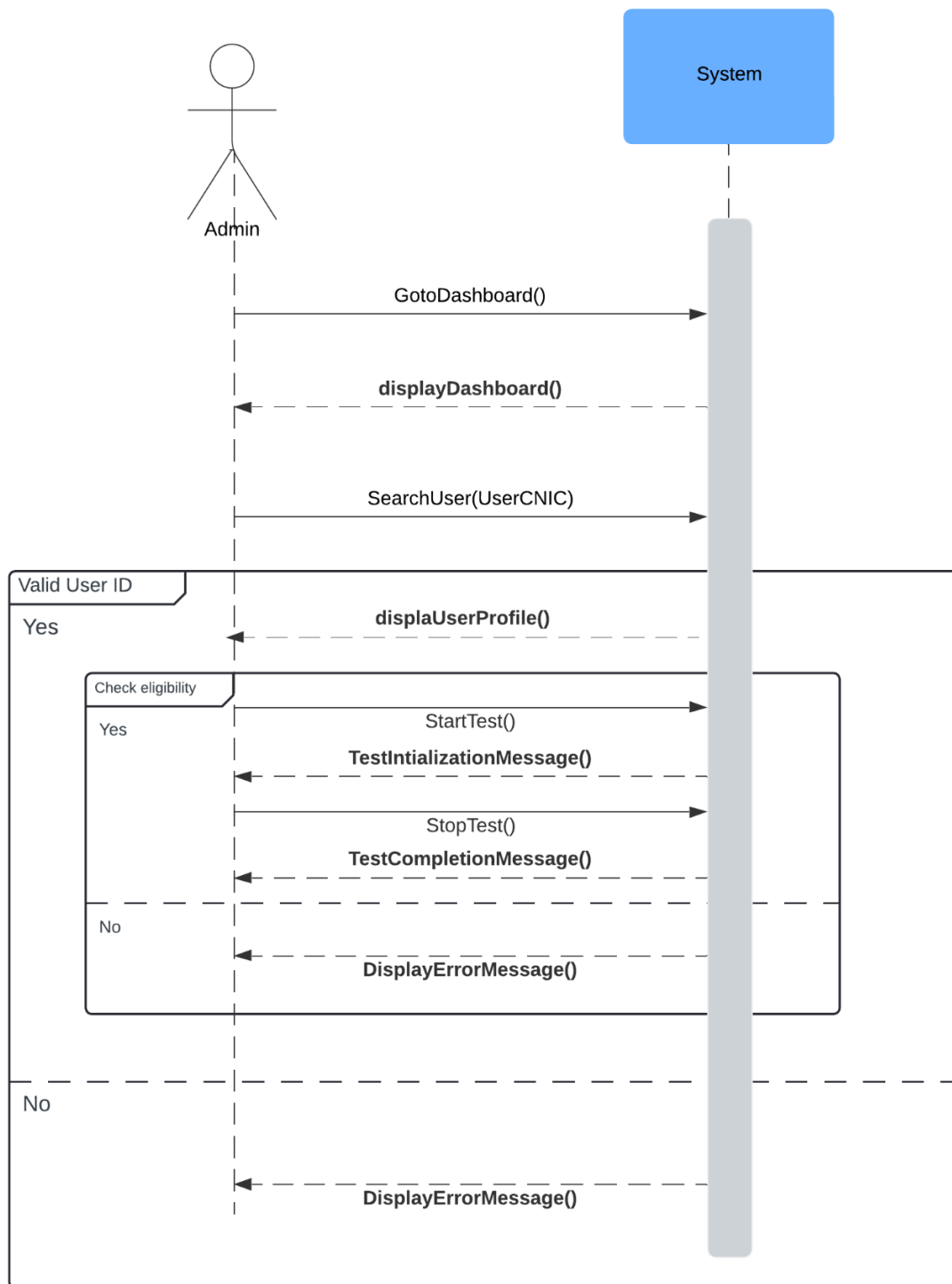


Fig 3.3.6 : Start/Stop Test System-level Sequence Diagram

Use Case # 07: Admin Review Feedback

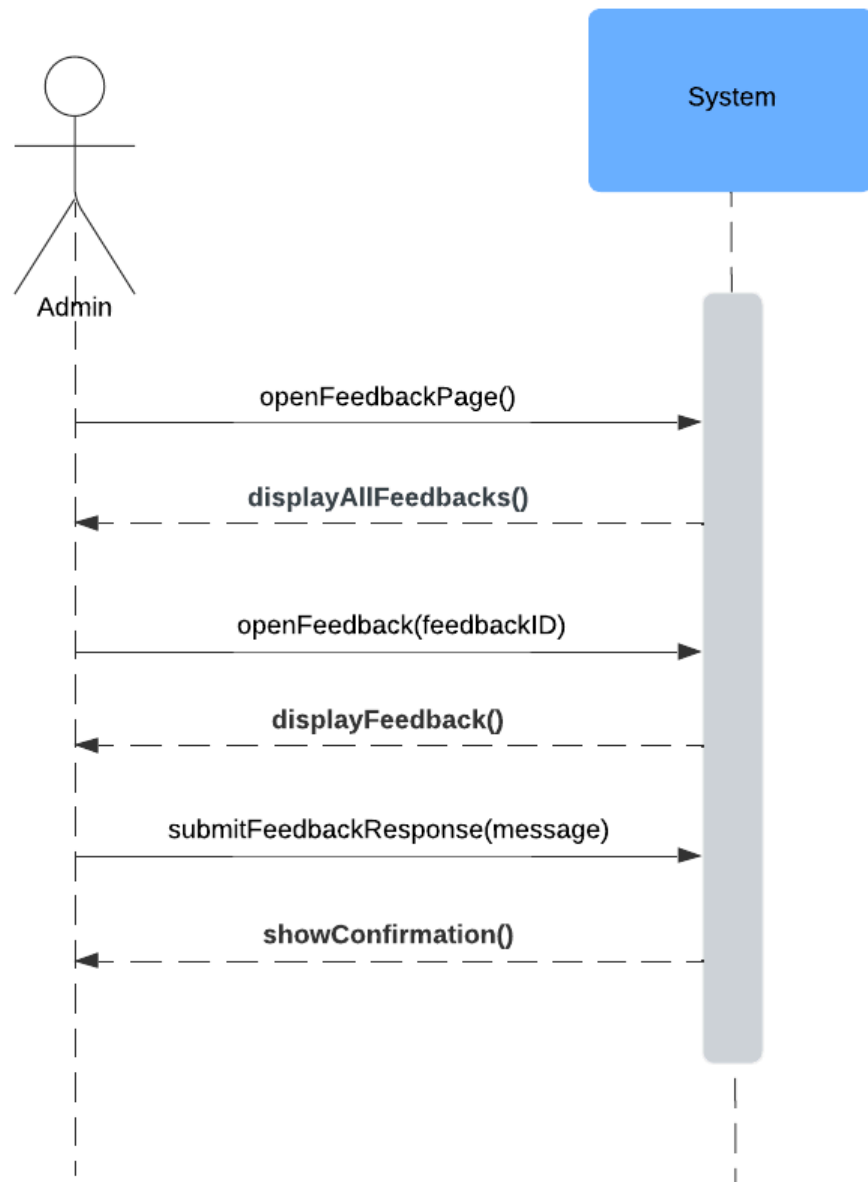
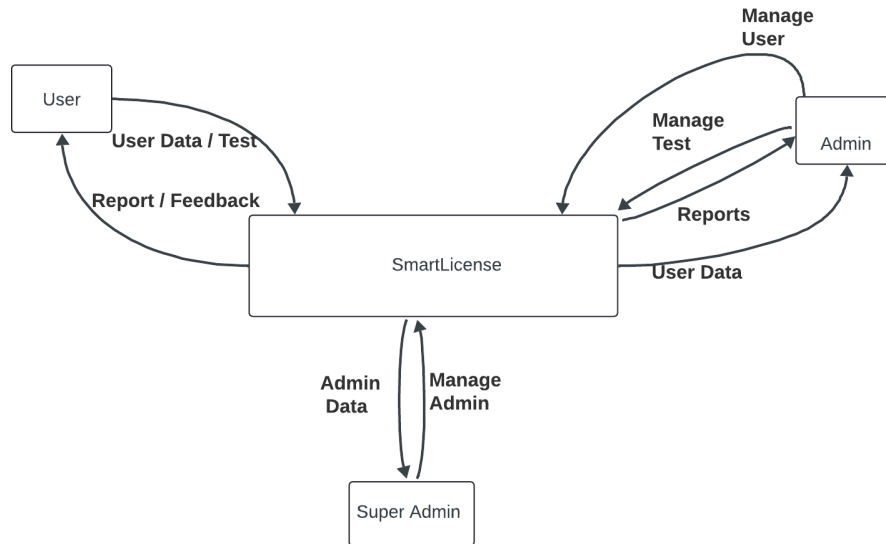


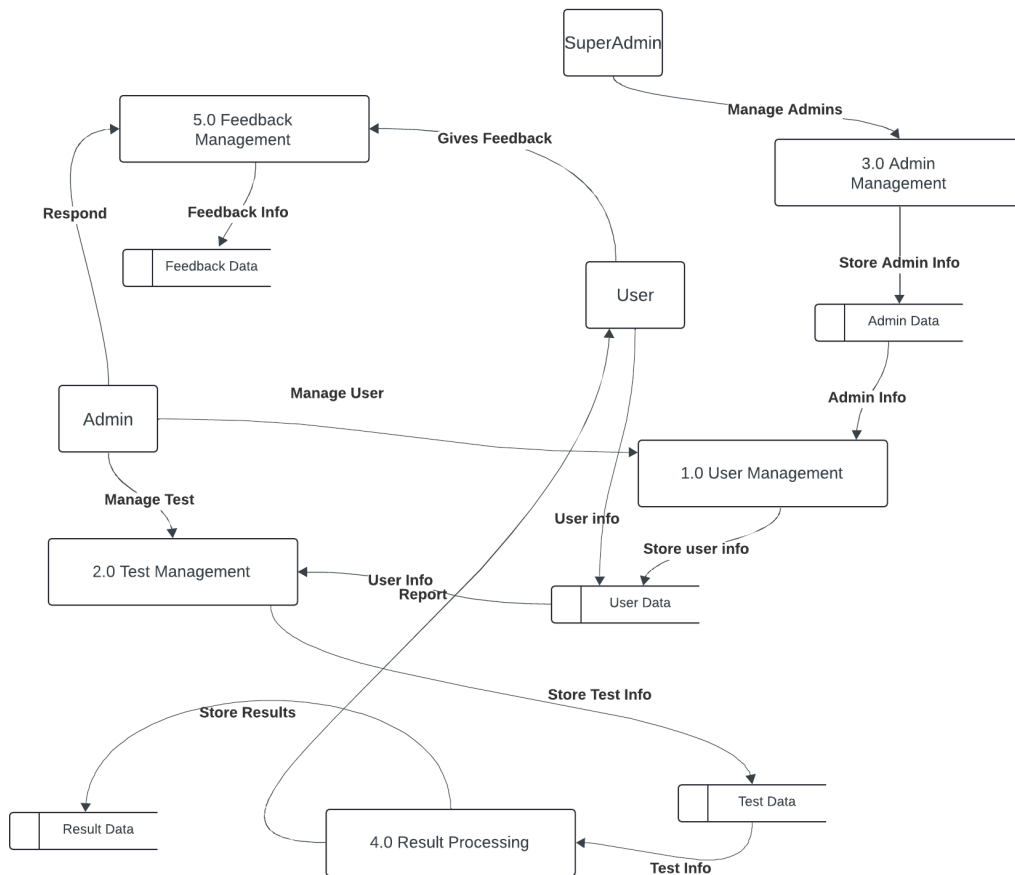
Fig 3.3.7 :Admin Review Feedback System-level Sequence Diagram

3.2.3 Data Flow Diagram:

3.2.3.1 Level 0



3.2.3.2 Level 1



3.2.3.3 Level 2

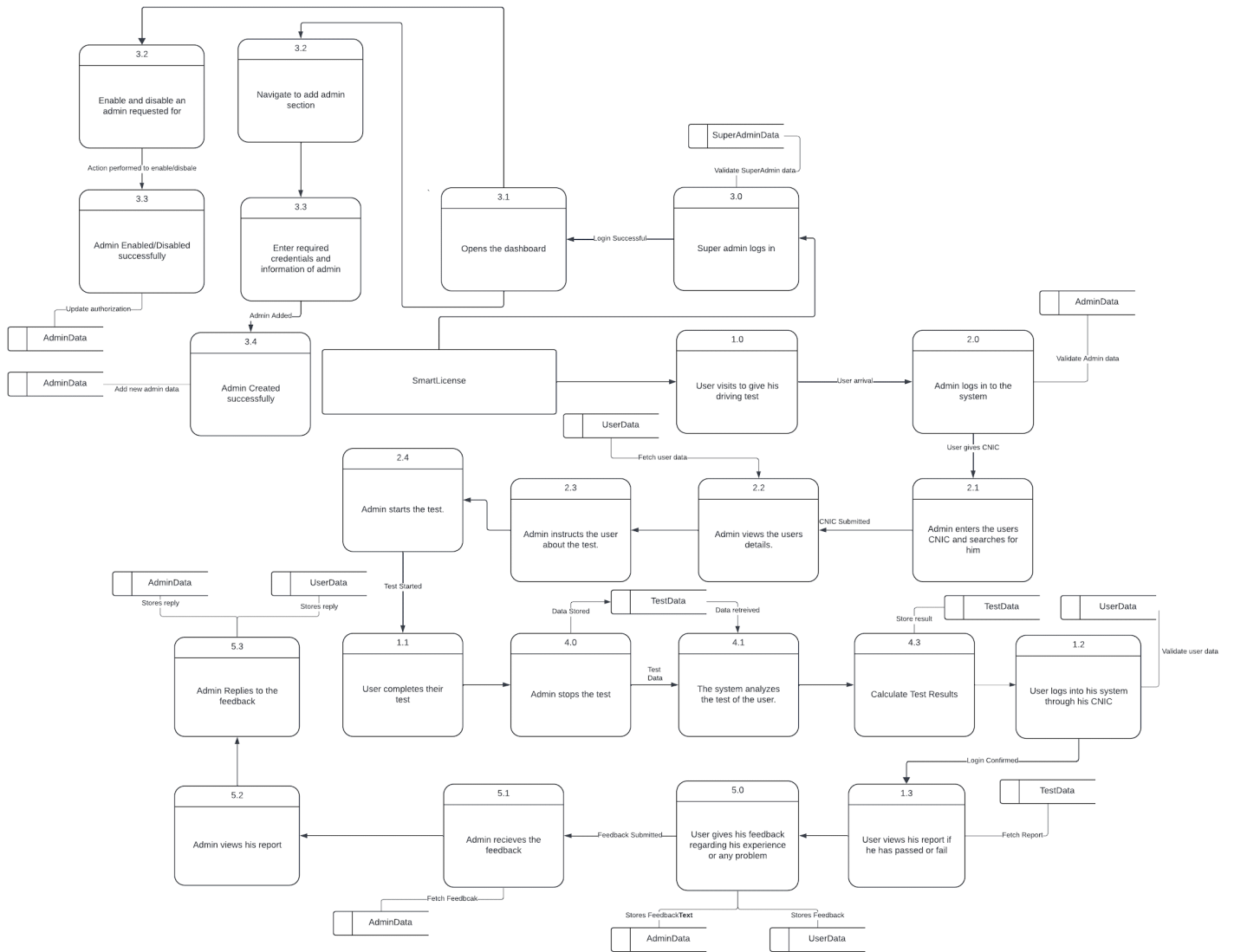


Fig 3.4: Data Flow Diagram

3.2.4 State Transition Diagrams:

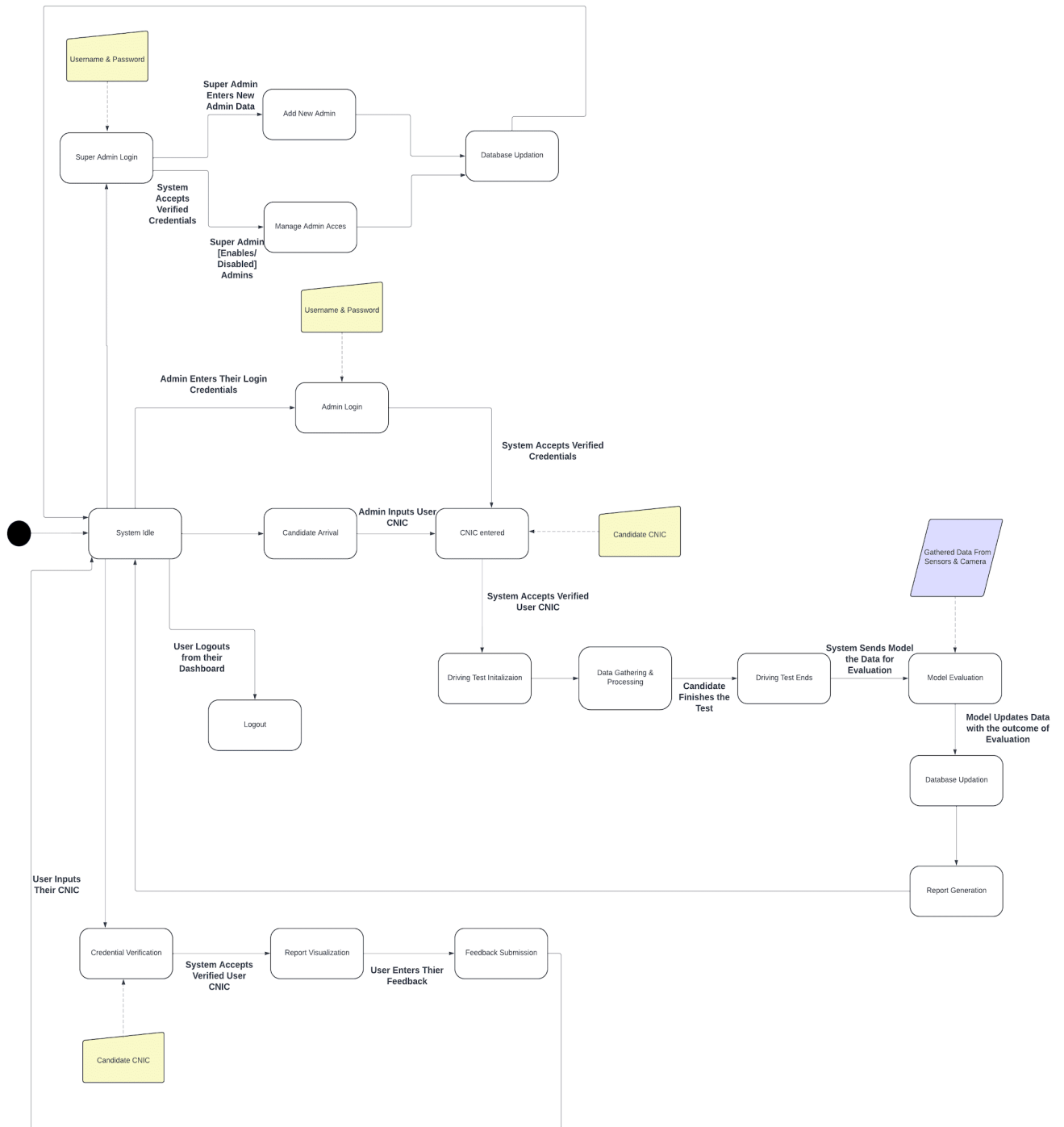


Fig 3.5 : System State Machine Diagram

3.3 Data Design

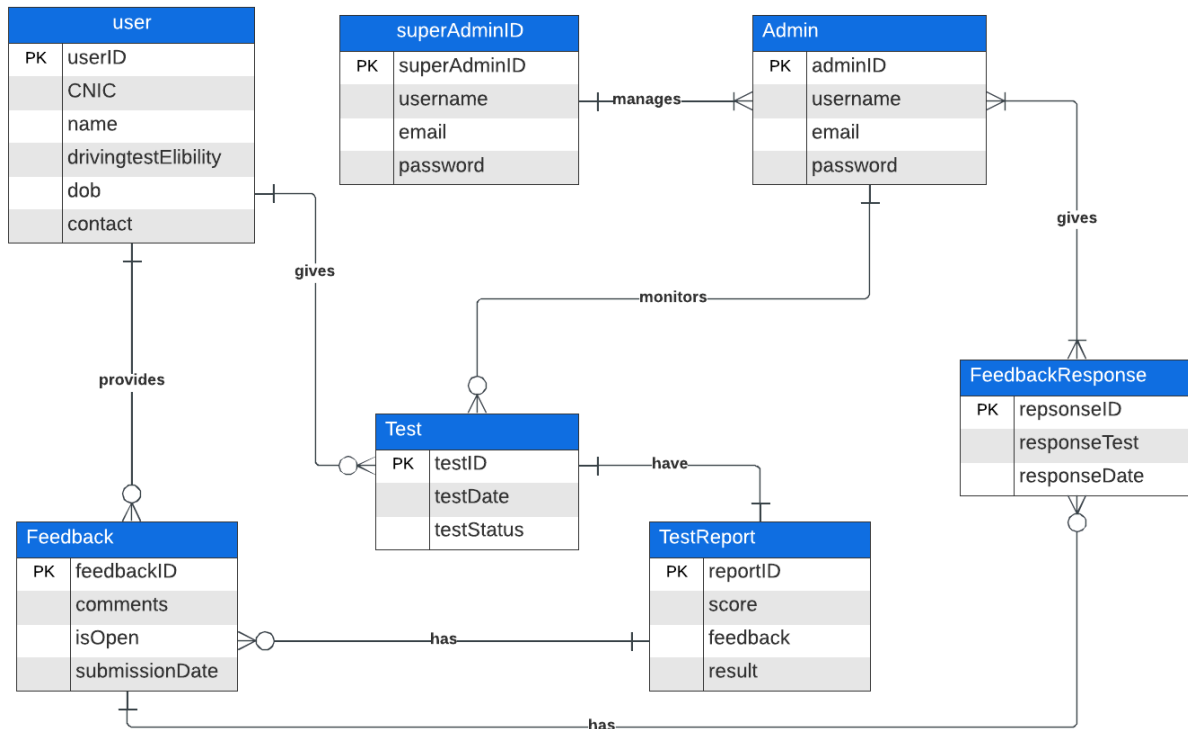


Fig 3.6 : Data Design Diagram

Data Transformation and Structure

1. Entities and Attributes:

- ➔ *User*: Maintains individual details and qualifications of a candidate.
- ➔ *SuperAdmin/Admin*: Oversees system participants and performs administrative activities.
- ➔ *Test*: Logs instances of a driving test along with its specific details.
- ➔ *TestReport*: Documents the results of tests, including scoring for each marking scheme.
- ➔ *Feedback/FeedbackResponse*: Manages candidate response over the test and score given whilst Admin can give Feedback response to each candidate.

2. Key Relationships:

Relationships like "manages", "monitors", and "provides" illustrates how various different roles interact within the system.

Data Organization and Management

1. **Database Structure:** Data is maintained in separate tables, each with a specific entity to ensure organized storage and retrieval of information. Each table has primary keys to uniquely identify each entry, e.g feedbackID and userID.

2. **Data Processing:** Basic CRUD operations help in effective management and accessing data. Foreign Key is used to maintain consistency across the tables, ensuring that related data is linked properly.

Data Storage Specifications

- **Database System:** The system uses a relational database, such as MySQL or PostgreSQL, to meet the requirement for the management of structured data and perform complex queries.
- **Security and Integrity:** Role-based authorization is implemented to distinguish between the access screens of SuperAdmin and Admin which improves the data security and ensures data integrity.

Chapter 4 - Poster

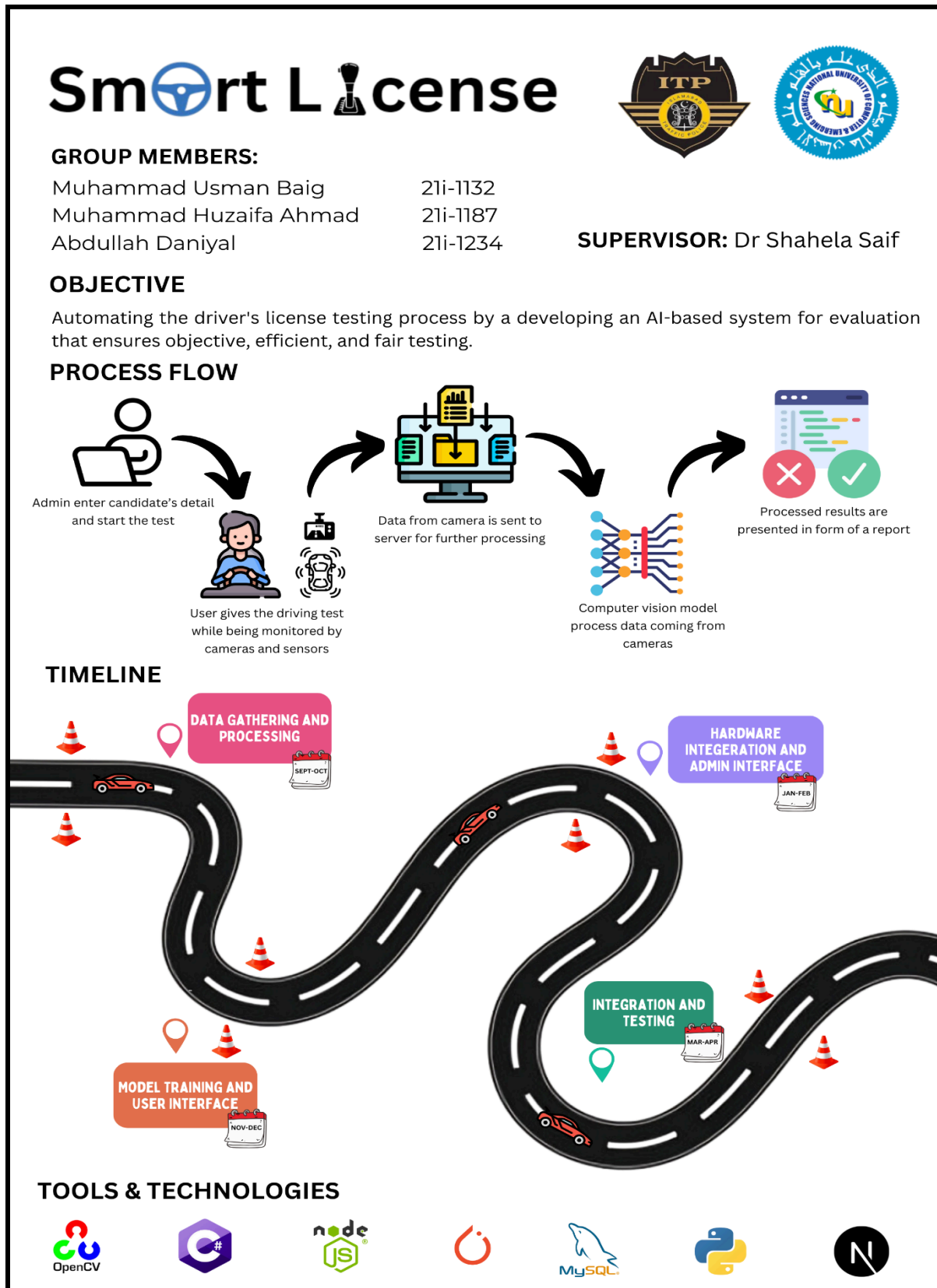


Fig 4.1: FYP Poster